

3.2 Algebra

3.2.1 Notation, vocabulary and manipulation

A1

Basic foundation content	Additional foundation content	Higher content only
• use and interpret algebraic notation, including: • ab in place of $a \times b$ • $3y$ in place of $y + y + y$ and $3 \times y$ • a^2 in place of $a \times a$, a^3 in place of $a \times a \times a$, a^2b in place of $a \times a \times b$ • $\frac{a}{b}$ in place of $a \div b$ • coefficients written as fractions rather than as decimals • brackets		

Notes: it is expected that answers will be given in their simplest form without an explicit instruction to do so.

A2

Basic foundation content	Additional foundation content	Higher content only
substitute numerical values into formulae and expressions, including scientific formulae		

Notes: unfamiliar formulae will be given in the question.

See the Appendix for a full list of the prescribed formulae. See also [A5](#)

A3

Basic foundation content	Additional foundation content	Higher content only
understand and use the concepts and vocabulary of expressions, equations, formulae, inequalities, terms and factors	to include identities	

Notes: this will be implicitly and explicitly assessed.

A4

Basic foundation content	Additional foundation content	Higher content only
simplify and manipulate algebraic expressions by:	simplify and manipulate algebraic expressions (including those involving surds) by:	simplify and manipulate algebraic expressions (including those involving surds and algebraic fractions) by:
- collecting like terms - multiplying a single term over a bracket - taking out common factors - simplifying expressions involving sums, products and powers, including the laws of indices		
	- expanding products of two binomials - factorising quadratic expressions of the form $x^2 + bx + c$, including the difference of two squares	- expanding products of two or more binomials - factorising quadratic expressions of the form $ax^2 + bx + c$

A5

Basic foundation content	Additional foundation content	Higher content only
understand and use standard mathematical formulae		
rearrange formulae to change the subject		

Notes: including use of formulae from other subjects in words and using symbols.

See the Appendix for a full list of the prescribed formulae. See also [A2](#)

A6

Basic foundation content	Additional foundation content	Higher content only
	know the difference between an equation and an identity	
	argue mathematically to show algebraic expressions are equivalent, and use algebra to support and construct arguments	to include proofs

A7

Basic foundation content	Additional foundation content	Higher content only
where appropriate, interpret simple expressions as functions with inputs and outputs		interpret the reverse process as the 'inverse function'
		interpret the succession of two functions as a 'composite function'

Notes: understanding and use of $f(x)$, $fg(x)$ and $f^{-1}(x)$ notation is expected at Higher tier.